SaaS Subscription Funnel & Churn

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Data Analytics Portfolio Project
October 2026

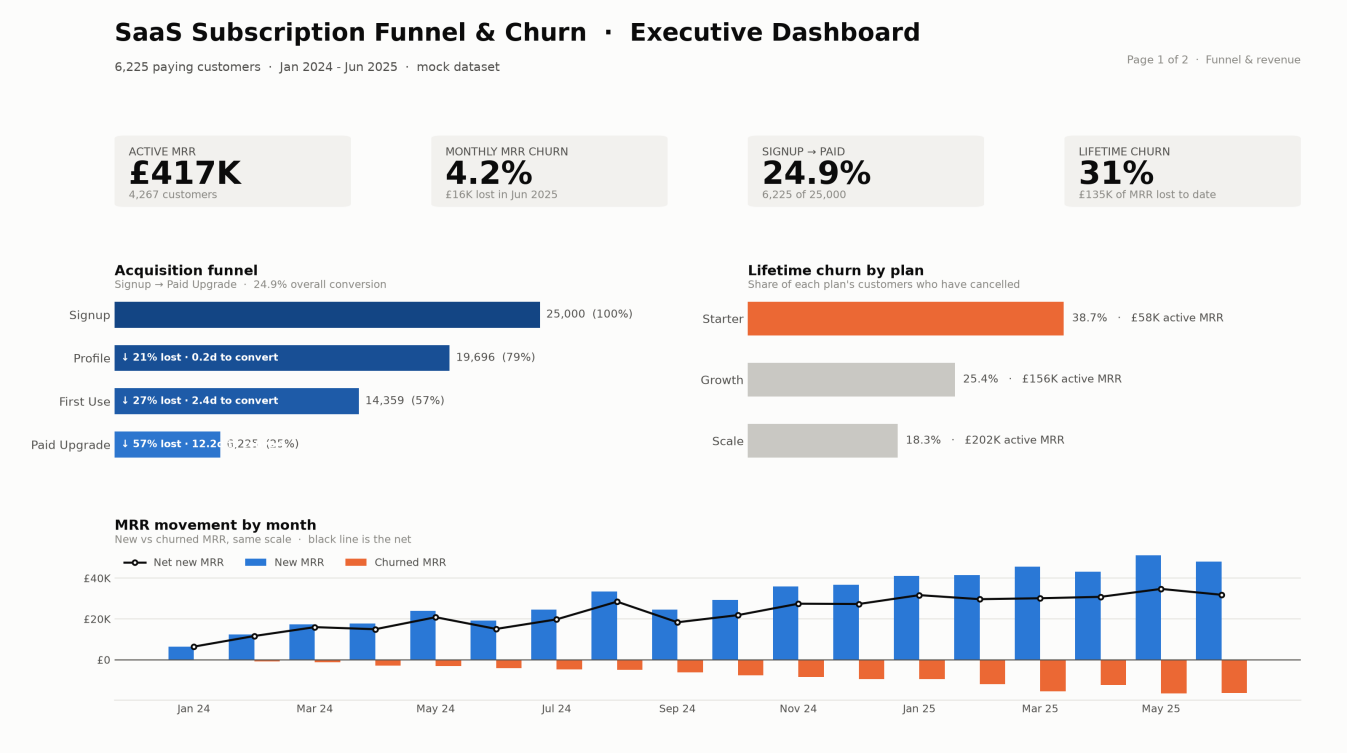
Funnel drop-off, cohort retention and MRR churn · Python · PostgreSQL · Power BI

Executive summary

A subscription business converts 25,000 signups into 6,225 paying customers and £417K of monthly recurring revenue — but it had no measurement of where signups stall, which cohorts survive, or what churn costs in revenue rather than logos. I built a pipeline that instruments the funnel from raw event logs, reconstructs each customer's active months in SQL, and reports retention and MRR movement on the same grain the finance team already uses.

Two findings change where effort goes. The funnel does not leak evenly: **57% of users are lost at a single step** (First Use → Paid Upgrade). And churn is front-loaded: **62% of all cancellations happen within three months** of signing up, which makes onboarding, not win-back, the lever.

ACTIVE MRR	MONTHLY MRR CHURN	SIGNUP TO PAID	MONTH-6 RETENTION
£417K	**4.2%**	**24.9%**	**64%**
4,267 customers	£16K in Jun 2025	6,225 of 25,000	weighted across cohorts



Dashboard page 1 — funnel drop-off, churn by plan and monthly MRR movement. Full dashboard: reports/saas_dashboard.pdf

What the analysis found

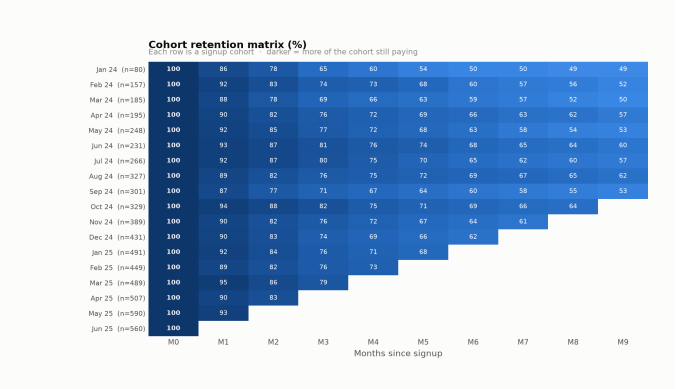
- **One step costs more than the rest combined.** First Use → Paid Upgrade loses 8,134 users (57%), and takes a median of 12 days — long enough for intent to decay.
- **Churn is an onboarding problem.** 1,217 of 1,958 cancellations (62%) happen in the first three months, taking £81K of MRR with them.
- **Price tier predicts survival.** Starter churns at 39% against Scale's 18% — but Scale holds £202K of active MRR, so the revenue risk and the logo risk sit on different plans.
- **Growth is outrunning churn, for now.** Net new MRR has stayed positive every month, but monthly MRR churn of 4.2% means roughly £196K a year has to be replaced before the business grows at all.

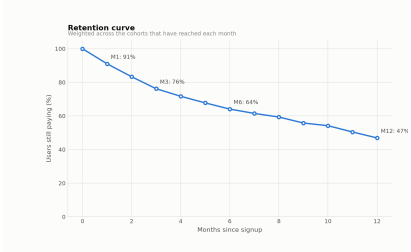
How it was built

Stage	Artifact	What it does
1. Funnel	**funnel_analysis.py**	Sorts raw events per user, collapses repeat emits to first touch, voids out-of-sequence stages, then reports absolute and relative drop-off plus median time-to-convert for each step.
2. Cohorts	**cohort_retention.sql**	PostgreSQL CTEs expand each customer into one row per active month with `generate_series`, then build the cohort retention matrix and a monthly new / churned / net MRR movement table.
3. Report	**power_bi_dax.txt**	DAX for churn rate, retention rate, active and lost MRR, net revenue retention and cohort retention, with the visual type each metric belongs in.

Modelling decisions worth noting

- **The retention denominator excludes cohorts too young to have reached the month.** A cohort that signed up last month has not lost its month-6 users; counting it as a zero understates month-6 retention by roughly half.
- **Churn rate is measured against the opening base.** Dividing this month's cancellations by this month's customer count flatters the rate whenever the business is growing, so the denominator is last month's close.
- **Out-of-sequence funnel events are voided, not counted.** A user appearing at Paid Upgrade with no First Use event is a logging artefact; left in, it silently inflates every conversion rate downstream.





Left: retention by signup cohort. Right: the pooled survival curve.

Recommended actions

Finding	Size	Recommended action
First Use → Paid Upgrade drop-off	8,134 users	Instrument the step, then trial in-product prompts at the median 12-day mark, before intent decays.
Churn in the first 3 months	1,217 customers	Move spend from win-back to onboarding; £81K of MRR is decided in that window.
Starter plan churn	39% lifetime	Test an annual option and a usage-based nudge into the mid tier.
Revenue concentration	£202K active MRR	Treat Scale accounts as named, not automated — each cancellation is worth ten Starter ones.

Where I would take it next

- **Split the funnel by acquisition channel.** A blended conversion rate hides which channels buy users who never activate.
- **Add expansion and contraction MRR.** Net revenue retention is the number a SaaS board asks for first, and upgrades are already in the event stream.
- **Model churn risk per account.** Tenure, plan and activation lag are enough for a first survival model.